National University of Computer and Emerging Sciences



AlchEx: A Non-Monetary Student Exchange Platform Based on Equivalent Exchange

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Objective:
The objective of **AlchEx** is to design and develop a cashless digital marketplace exclusively for students, where goods and services are exchanged without the use of money. The platform is inspired by the alchemical principle of Equivalent Exchange, which states that ***humankind cannot gain anything without giving something in return***. AlchEx aims to encourage collaboration, skill-sharing, and mutual support among students by enabling fair and transparent exchanges within a trusted academic environment

Scope:

AlchEx will function as an exchange ecosystem with the following core features::

1. **User Authentication and Access Control:**
   Students will be able to register and log in using verified .edu email addresses only, ensuring that the platform remains exclusive to students. There will be two types of users in the system: Students and Admins. Secure authentication and account management will be implemented for all users.

2. **Student Marketplace (Requests and Offers):**
   Students will be able to create posts describing a request for a service or good, along with what they are willing to offer in return. Examples include academic help, sharing course resources, or small everyday items such as coffee or snacks. Other students can respond to a post by accepting the listed offer or proposing an alternative exchange. Each student can submit only one offer per post, while a student may create multiple requests simultaneously.

3. **Post Lifecycle and Status Management**:
   Each post will have a defined active time period, after which it will automatically expire and be removed from the active listings. Every post will display a clear status such as Active, Accepted/In Progress, Successful, or Expired. By default, only active posts will appear on the main feed, while other statuses will be accessible through filters and user history.

4. **Acceptance, Communication, and Transaction Completion:**
   The original requester will have full control to accept or reject incoming offers. Once an offer is accepted, a private chat will be enabled between the two students to coordinate the exchange. After the exchange is completed, both users can mark the transaction as successful. During the in-progress stage, either participant may choose to cancel the transaction before completion, after which the post status will be updated accordingly.

5. **Ratings, Reputation, and Leaderboard:**
   After a successful exchange, students will be able to rate each other based on their experience. These ratings will contribute to a reputation score visible on user profiles. A

leaderboard will be included to highlight top contributors based on successful transactions and ratings, with filters such as Today, Last Week, and Last Month.

6. **User Profiles and Filtering:**
   Each student will have a profile displaying a profile picture and basic academic information, including name, email, batch, program, section, university, and campus. Students may also add a short personal description and optional social media links. The platform will provide filters and sorting options based on university, campus, batch, program, and section to help users find relevant exchanges easily.

7. **Admin Panel and Moderation:**
   Admins will have a dedicated dashboard with moderation capabilities. They will be able to monitor posts, manage users, resolve disputes, remove inappropriate listings, and ban users if necessary. Admins will also have the authority to contact users directly for warnings or clarification in case of policy violations.

8. **Reporting and Dispute Handling:**
   Students will be able to report a transaction if a mishap occurs, such as failure to deliver the promised service or good. Reported cases will be visible to admins, who can review the dispute and take appropriate action.

9. **AI-Assisted Moderation (Optional):**
   If time permits, AI-based tools may be integrated to assist with content moderation by detecting spam, inappropriate content, or suspicious behavior, helping admins maintain a safe and fair environment.

Conclusion:
AlchEx proposes a unique, non-monetary exchange system tailored specifically for students, promoting fairness, collaboration, and community engagement. By combining structured moderation, transparent transactions, and the philosophy of equivalent exchange, the platform aims to create a trustworthy and engaging student-driven marketplace within the academic ecosystem.